

Customization Quick Reference — Precision Flow Conditioner

Expected Effects of Key Parameter Changes (Qualitative — Validate Experimentally)

Parameter	Direction	Main Effect	Secondary Effects	Notes
Gap	Smaller	↑ Velocity & shear	Higher Re, smaller bubbles	Harder to print
Gap	Larger	Gentler flow	Lower Re, larger bubbles	Easiest to build
Nozzle °	Shallower (6-8°)	Stronger spiral	Better conditioning, ↑ΔP	Best uniformity
Nozzle °	Steeper (12-15°)	Easier flow	Less conditioning	High throughput
HVAT Brake	30-50% locked	Major swirl reduction	Clean axial jet	Best tuning knob
# Discs	More (30-35+)	More channels	Smoother, taller device	More time/material
Orifice	Smaller (6mm)	Highest velocity	Most violent collapse	Erosion risk

DISCLAIMER: Qualitative expectations from simplified models. Real results depend on surface finish, alignment, inlet quality, fluid properties, etc. Always prototype and measure.